TSI HIRING PACKAGE

2025 - 2026

DEVELOPER PACKAGE

OPENING STATEMENT

At most universities, tech is taught as a skill, something to list on a resume, use in an internship, or build a product with. But what often gets left out is the "why" behind the work, who it's for, who it serves, and the changes it create.

At TSI, we believe technology is most powerful when it serves a purpose beyond itself. We're here to prove that tech can be purpose-driven, community-rooted, and human-first.

We bridge the gap between student talent and real-world social needs. Through hands-on nonprofit projects and cross-functional collaboration, our members apply their skills to work that truly matters.

Whether you're breaking into tech or searching for more meaningful work, TSI is a space to grow with purpose. We believe the most fulfilling careers are built at the intersection of skill, curiosity, and impact.

Here, you'll gain hands-on experience, mentorship, and a community that challenges you to think not just what you want to do, but why you want to do it. Whether you're here to explore, to build, or to lead, this is where your journey begins.

Welcome to TSI!
David Liu

DEVELOPER

This role is about growth. We don't expect you to come in with all the answers, in fact, **no experience** is required. What matters is your willingness to learn, to show up, and to care about what you're building. You'll be paired with other students, mentored by experienced PMs, and supported through every step of your journey. You'll commit to a real project, contribute to a real team, and leave with something tangible.

RESPONSIBILITY

- **Contributing to real, mission-driven software** – You'll help build tools used by nonprofits, from web apps to data systems, using modern technologies alongside a dedicated team.
- **Asking questions and seeking feedback** – You're not expected to know everything, but you are expected to learn. Growth is part of the job.
- **Participating in weekly meetings and team updates** – Communication is key in a team project. You'll keep your team in the loop and stay aligned on goals.
- **Learning fast, failing forward, and iterating often** – Projects evolve. Problems pop up. You'll adapt, find solutions, and move forward — just like in the real world.
- **Building things that actually matter** – This isn't a side project. The work you do here will end up in the hands of real organizations solving real-world problems.

THE PERFECT CANDIDATE

- **Ambition**: You're hungry to learn and eager to grow
- **Consistency**: You show up, follow through, and communicate
- **Team Spirit**: You collaborate, give feedback, and support others
- **Impact-Driven Mindset**: You want to build things that matter

Experience is not required.
We welcome students from all backgrounds, whether you're in CS, engineering, science, business, or just love building things.
If you're curious, committed, and excited about using tech to create social change. We want you on the team.

PM Noah Kostesku
Thomson Lam

WORLD VISION

World Vision is a global humanitarian organization focused on improving the lives of vulnerable communities through sustainable development, emergency relief, and advocacy. They have operations in over 150 countries, and this project supports their innovation team by accelerating the idea discovery pipeline, which includes optimizing their process of identifying high-impact, fundable, and scalable initiatives to deliver wins for the organization faster.

AGENTIC AI DISCOVERY PIPELINE

The project introduces a modular set of AI agents into World Vision’s discovery process. These agents will handle repetitive research, criteria grading, and financial sanity checks, while keeping human reviewers in control of approvals. We will be doing bi-weekly sprints to develop an initial POC (Proof of Concept) to demo to World Vision, then optimize and deploy a professional solution in the later stages of the project. The MVP will:

- Use Airtable as the intake and review backlog
- Deploy onto Azure for orchestration
- Connect LLMs (hosted or open source) for reasoning and scoring
- Store embeddings and transcripts in a vector DB + some kind of storage
- Provide an auditable, Human-In-The-Loop review process

TECH STACK

TBA: dependent on client preferences (existing codebase or Microsoft ecosystem).
Be prepared to be exposed to the following tech:

- **Languages:** Typescript, Python
- **Frontend:** React/NextJS, TailwindCSS, Vercel AI SDK
- **Agentic Frameworks:** LangChain/LangGraph, Langsmith/Langfuse, FastAPI
- **Bonus Skills:** LLMs, Vector Databases (HuggingFace, Pinecone, Azure Cognitive Search), benchmarking agents (AgentBench), Azure (Functions, Blob Storage, Power Automate, Monitor), API security, Azure AD, PII scrubbing middleware

PROJECT DIFFICULTY

4/5 – difficulty mostly lies in benchmarking performance of a multi-agent system (arbitrary metrics such as the definition of a “good” idea), and deployment + optimization on Azure.

PM Andres Pedreros
Sarah Chiang

INTERNATIONAL RESCUE COMMITTEE

The International Rescue Committee (IRC) responds to the world's worst humanitarian crises, helping to restore health, safety, education, economic wellbeing, and power to people devastated by conflict and disaster. Operating in over 40 countries and 28 U.S. cities, IRC helps individuals survive, recover, and rebuild their lives.

SETTLE IN MOBILE APP

Settle In is a mobile and web application designed to support newcomers throughout their resettlement journey. Acting as a "go-to companion" for refugees and immigrants, the app provides multilingual cultural orientation through short videos, interactive lessons, and progress tracking features. Due to changes in the U.S. Refugee Admissions Program and a lack of funding, the app's content has become outdated and costly to maintain. IRC is seeking a partner to:

- Rebuild the mobile application
- Update content to align with the current Settle In website
- Implement a cost-effective and sustainable architecture
- Ensure multilingual, interactive, and accessible features for newcomers

TECH STACK

TBA: determined by IRC based on their final implementation direction.
Be prepared to be exposed to the following tech:

- **Languages & Frameworks:** JavaScript, TypeScript, React Native, Flutter, Expo
- **State Management:** Redux, MobX, Provider, Riverpod, Bloc
- **API Integration:** RESTful API, GraphQL
- **UI/UX:** Responsive design, React Native Paper, NativeBase, Material UI, TailwindCSS (with RN), Lottie, React Native Reanimated, Flutter animations
- Accessibility & Localization: i18next, react-intl, Flutter intl, RTL language support
- **Infrastructure:** Git/GitHub/GitLab workflows
- **Notifications & Analytics:** Firebase Cloud Messaging, OneSignal, Firebase Analytics, Amplitude, Mixpanel
- **Authentication:** OAuth2, Firebase Auth, Auth0 SDKs

PROJECT DIFFICULTY

4/5 – high client expectations for polish, reliability, and sustainability. Requires multilingual support (including RTL), cultural sensitivity in UX, cost-effective infrastructure, and long-term maintainability.

PM Omar Hossain
Alan Sanjit

LONDON CHILDREN MUSEUM

The London Children's Museum has been providing a space for children to learn through play-based learning that fosters cognitive, social, emotional, and physical development for over 45 years. Founded in 1975, the museum now welcomes over 200,000 children and families each year and showcases more than 7,000 artifacts that allow children to explore history, science, social relationships, and art.

ASTRONAUT TASK INTERACTIVE

As part of their "Above & Beyond" exhibit, the London Children's Museum has partnered with us to develop an Astronaut Glove Interactive Exhibit. This hands-on installation will simulate the challenges astronauts face when working in space. Visitors will wear a custom-designed astronaut glove equipped with sensors and use a joystick to complete repair tasks on screen, providing a tactile and immersive experience of working in zero gravity.

This interdisciplinary project involves three integrated teams – mechanical, electrical, and software – and will require full-stack development of the physical hardware, game environment, and enclosure design. The final product will be museum-grade in durability, accessibility, and visual appeal.

TECH STACK

Be prepared to be exposed to the following tech:

- **Game Engine:** Unity
- **Programming Language:** C#
- **3D Modelling:** Blender
- **Hardware Platform:** Arduino, Teensy, ESP32
- **Sensors**: Flex sensors, gyroscope, accelerometer (MPU6050)
- **Input Devices:** USB joystick, custom astronaut glove
- **Enclosure Design:** CAD modeling, mechanical fabrication

PROJECT DIFFICULTY

3.5/5 – While the individual components are manageable on their own, the primary challenge is in integrating all three disciplines into a final product. In particular, coordination between the electrical team's sensor systems and the software team's game environment will require careful collaboration, testing, and refinement.

PM Lucian Lavrict
David Liu

CHILDCAN

Childcan is a non-profit charity supporting families whose children are undergoing cancer treatment. These treatments are often emotionally and financially overwhelming, and Childcan provides support throughout the entire journey: including treatment, recovery, and bereavement. The organization relies entirely on donations and external support, making donor relationships critical to its mission and continued impact.

WEBSITE REDESIGN

Childcan's current website is fragmented across Wix and Squarespace, limiting its scalability, customization, and overall user experience. This project will rebuild and unify the entire website using a custom React application to better serve families, donors, and supporters. The rebuild will:

- Unify the platform: Consolidate all content into a single, maintainable React application
- Enhance performance: Improve speed, accessibility, and responsive design
- Enable customization: Offer control over design, layout, and interactive features like events, donations, and family stories
- Streamline updates: Allow staff to manage and update content easily while maintaining a cohesive brand identity

The goal is to ensure the site reflects Childcan's mission while offering a modern, accessible, and user-friendly digital presence that aligns with their growing impact.

TECH STACK

Be prepared to be exposed to the following tech:

- **Languages & Frameworks: JavaScript, React, Node.js**

PROJECT DIFFICULTY

2/5 – the tech stack is simple and familiar, but challenges may arise around deployment, scalability, and long-term maintainability.

PM Marco Chen
Andrew Durnford

PLAN INTERNATIONAL

Plan International is a global humanitarian and development organization advancing children's rights and equality for girls. Founded in 1937, it operates in over 75 countries, addressing the root causes of challenges faced by children and their communities through advocacy, education, and sustainable development.

FRF SYSTEM MODERNIZATION

Plan International is seeking to modernize its critical Financial Revenue Framework (FRF) system, a legacy Visual Basic application used to process expense data from global offices for financial reporting. The current system:

- Ingests CSV data from SAP and Great Plains
- Applies hardcoded mapping logic to convert expenses into revenue accounts
- Provides reconciliation features for month-end closing
- Exports data to financial systems

Key issues include inefficient line-by-line processing, nested if-statements for mapping logic, poor maintainability, and lack of documentation. This project will:

- Replace hardcoded logic with database-driven mapping tables
- Use SQL-based batch processing for efficiency
- Refactor or rebuild the system using modern technologies
- Maintain compatibility with the existing SQL Server database

The result will be a maintainable, performant system aligned with the organization's growing financial reporting needs.

TECH STACK

Data Processing: Python, Pandas
Database: SQL Server, SQLAlchemy ORM
Backend: FastAPI
Frontend: React or FastAPI templates (if reconciliation UI is needed)

PROJECT DIFFICULTY

3/5 – the primary challenges involve reverse-engineering undocumented VB logic, understanding embedded financial rules, and ensuring data integrity during migration. However, inputs/outputs are well defined, and the team will have support from stakeholders familiar with the system.

PM Thomas Llamzon
Anthony Lam

PLAN CATALYST

PlanCatalyst is a consulting subsidiary of Plan International, focused on international development and corporate sustainability. With expertise in gender equality, climate resilience, education, health, economic empowerment, and impact measurement, PlanCatalyst supports organizations in driving meaningful, measurable progress toward sustainable development goals.

GLOBAL DEVELOPMENT DATA DASHBOARD

PlanCatalyst is developing a public-facing, interactive data dashboard to be embedded in their redesigned website. This dashboard will visualize country-level indicators across key development themes, including:

- SDG progress
- Demographic trends
- Climate vulnerability (ND-GAIN)
- Financial inclusion
- Governance
- Gender equity

The dashboard will serve as a showcase of PlanCatalyst's analytical capabilities and provide stakeholders with localized, policy-relevant data. Key features include:

- Integration of data from publicly available APIs (World Bank, UN, etc.)
- Development of a composite index to rank countries across selected indicators
- Interactive geographic navigation and filtering by region or theme
- Support for comparative visualizations
- Scheduled or ad hoc data refresh capabilities
- Seamless embedding into PlanCatalyst's Wix-hosted website

TECH STACK

Data Processing: Python (Pandas, GeoPandas – optionally)
Data Gathering: Public APIs (World Bank, UN datasets)
Data Visualization: Tableau, Microsoft Power BI
Automation: GitHub Actions, AWS Lambda
Storage: AWS S3, Microsoft OneDrive, Google Drive

PROJECT DIFFICULTY

3/5 in terms of technical implementation but expectations for polish, reliability, and flawless delivery elevate the difficulty to a 4/5. Strong collaboration and consistent communication are essential.

PM Jack Hogan
Jack Ruttan

CANADIAN RED CROSS

The Canadian Red Cross (CRC) is one of Canada's largest and most trusted humanitarian organizations, providing life-saving support and essential services across the country and globally. With a legacy of disaster relief, emergency response, and health education, CRC plays a critical role in Canada's national emergency infrastructure. In 2024 alone, the organization supported over 7.8 million people and raised more than $164 million to fund domestic and international humanitarian operations, making it one of the most impactful non-profits in the country.

PAVED ROADS DEVELOPER PLATFORM

This project focuses on creating a standardized internal developer platform to improve how Red Cross engineering teams build, test, and deploy software. The goal is to reduce infrastructure overhead, ensure security compliance, and increase delivery speed through reusable pipelines and observability tooling. Deliverables include:

- Secure CI/CD pipeline templates for web and API services
- Default observability dashboards for error tracking, uptime, and performance
- Automation for collecting software delivery metrics (DORA metrics)
- Streamlined app review process with checklists and automation for ARB (Architecture Review Board) packets

This initiative will reduce lead time, improve software quality, and promote consistent engineering standards across internal teams.

TECH STACK

CI/CD: GitHub Actions, Azure DevOps
Observability: Prometheus, Grafana, Datadog
Containerization: Docker, Kubernetes
Languages: Java, Go, Shell scripting

PROJECT DIFFICULTY

4/5 – requires infrastructure-level thinking, cross-functional collaboration, and integration with CRC's internal DevOps systems.